

Multimodal Integration of Visual Masking and Phantom Sensation for 3D Object Interaction in Hand Projection

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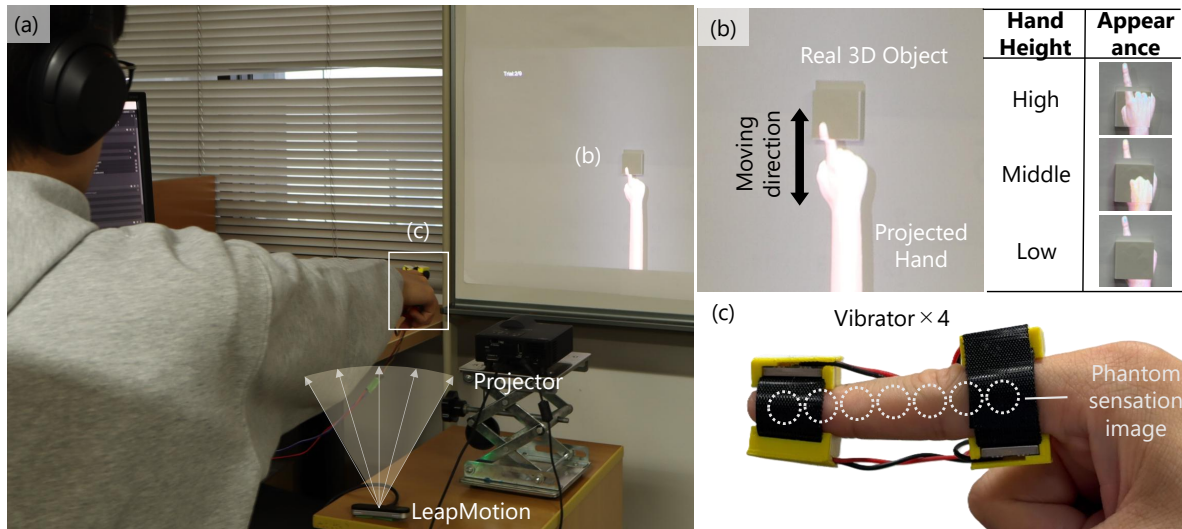


Figure 1: Wearable Haptic Feedback and Visual Effects in Hand Projection onto 3D Objects. (a) System Overview. (b) Visual effects change based on real hand height. (c) Wearable Haptic Feedback and Vibration Presentation Based on Projection Position.

ABSTRACT

Projected hand-arm extension technology (ExtendedHand) enables users to interact with remote objects beyond their physical reach, but faces challenges such as visual discrepancy from 2D projection on 3D objects and a lack of haptic feedback, which inhibit bodily perception. This study proposes a multimodal presentation method that integrates dynamic visual masking for depth-aware spatial perception and position-dependent vibration using Phantom Sensation (PhS). Our user study revealed modality-specific effects: visual factors primarily contribute to the sense of agency and penetration sensation, while vibration independently enhances tactile realism. In particular, the physically unnatural penetration effect serves as a symbolic feedback mechanism that maintains operational continuity, leading to a significantly higher sense of agency compared to the non-penetration condition. In conclusion, the combination of the penetration effect and position-dependent vibration provides an optimal configuration for achieving a heightened penetration sensation and practical usability without compromising the bodily experience of the projected hand.

Index Terms: Spatial augmented reality, human augmentation, projected hand, phantom sensation.

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1 INTRODUCTION

In recent years, driven by advancements in artificial intelligence and robotics, human augmentation technologies that extend physical and cognitive capabilities have garnered significant attention [6]. One such example is ExtendedHand, proposed by Ueda et al. [10], which employs a projection-based system to render a hand at a distance, enabling intuitive interaction with areas beyond the user's physical reach.

Building upon the initial concept of ExtendedHand, Hanai et al. achieved high portability by developing a wrist-mounted device that allows users to perceive the vibration characteristics of remote objects [2]. Additionally, Sato et al. improved interaction quality by dynamically rendering shadows for the projected hand, thereby visually distinguishing between “hovering” and “contact” states—a distinction that was previously challenging due to the inherent constraints of projection technology [8].

However, current systems typically assume that the projection target is a planar surface, imposing significant restrictions on depth-wise interactions with real 3D objects, such as “moving behind” or “passing through” them. Due to technological constraints, conventional projection methods cause the extended hand to be rendered exclusively on the object's surface, effectively locking its depth-wise spatial relationship with the object.

In contrast, interaction research in virtual reality (VR) suggests that intentionally allowing visual interpenetration—where the hand appears to sink into an object—can optimize usability beyond real-world physics, such as by enhancing shape perception or simplifying complex grasping tasks [5]. This implies that leveraging non-realistic VR behaviors as symbolic feedback may extend human bodily perception more effectively than strictly adhering to physical consistency.

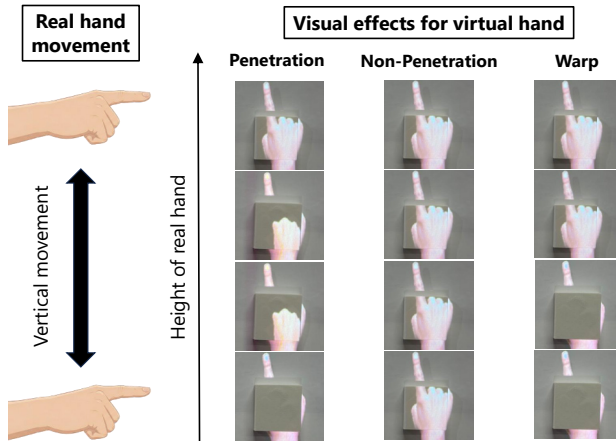


Figure 2: Three types of visual effects.

In this study, we propose implementing this VR-like flexibility in physical space using a projection-based system to recreate non-realistic behaviors, such as a hand passing through an object, as shown in Fig. 1. Specifically, we introduce visual effects such as “penetration, non-penetration, and warping” by applying dynamic masking to an extended hand controlled via hand tracking, accounting for the 3D geometry of the objects. Simultaneously, we integrate Phantom Sensation (PhS), which controls the amplitude ratio of multiple vibrators to present a continuous sensation of movement as if tracing across a finger.

PhS is a tactile illusion where a single stimulus is perceived between two distant points on the skin; since its initial report as an information transmission method by Alles [1], it has been applied to various haptic displays. Furthermore, it has been reported that PhS can be induced within the body [4], a finding that can be applied to the non-realistic sensation of an object penetrating the body. It has also been shown that motion perception via PhS is strongly influenced by visual feedback [7].

Based on the integration of these visual transformations and tactile stimuli, we aim to clarify their effects on the user’s bodily perception. To this end, we formulated the following hypotheses regarding the sense of agency and tactile realism:

- **H1 (Visual):** Visual effects that allow penetration (Penetration/Warp) will elicit a higher sense of agency and body ownership than the physically correct Non-penetration effect, by resolving the spatial conflict between motor intention and visual feedback.
- **H2 (Haptic):** Position-dependent vibration using PhS will elicit higher tactile realism than synchronous vibration or no vibration, by providing spatial cues consistent with the visual depth.

2 PROPOSED METHOD

The overview of the proposed system is shown in Fig. 1(a). This system facilitates interaction with 3D objects placed on a plane in physical space by using a projection-based virtual hand controlled via hand tracking. By combining projected images with a wearable vibration device, the system is characterized by its ability to modulate the visual depth-based spatial relationship relative to objects (Fig. 1(b)) and provide haptic feedback with a sensation of movement synchronized with the spatial interaction between the virtual hand and the 3D objects (Fig. 1(c)).

2.1 Visual Effects

As shown in Fig. 1(b), the appearance of the projected virtual hand is dynamically transformed by applying a masking process to the projection area corresponding to the 3D object. While conventional projection methods are limited by the hand always being rendered atop the object’s surface, our method represents the depth-wise position relative to the object by selectively occluding parts of the hand through masking.

In our Unity-based implementation, virtual objects with coordinates and geometries corresponding to the physical 3D objects are aligned within the virtual space. We adopted a method to dynamically control the rendering of the virtual hand by applying occlusion shaders and masking to these virtual objects. Specifically, as illustrated in Fig. 2, the system switches between the following three visual conditions based on the user’s real-hand height data acquired via a Leap Motion controller:

- **Penetration effect:** The mask’s range changes in conjunction with the vertical movement of the real hand, creating the illusion of the virtual hand passing through the object.
- **Non-penetration effect:** A baseline condition where the virtual hand is consistently rendered on the object’s surface regardless of the real hand’s height.
- **Warp effect:** The occlusion state of the virtual hand changes abruptly and discontinuously when the real hand exceeds a specific height threshold.

These effects provide a non-realistic interpenetration experience within physical space, similar to those found in virtual reality.

2.2 Vibration Effects

To provide haptic feedback synchronized with the visual depth, we utilize PhS. PhS is a tactile illusion where stimulating two distant points on the skin simultaneously allows the user to perceive a single virtual vibration source between them [1]. As shown in Fig. 1(c), the system utilizes a device equipped with four Linear Resonant Actuators placed on the index finger: at the tip and base of both the palmar and dorsal sides.

We implemented two vibration modes to investigate the effect of spatial cues:

- **Position-dependent vibration:** This mode utilizes PhS to simulate the sensation of the object passing through the finger. By dynamically controlling the amplitude ratio between the tip and base vibrators according to the penetration depth, the perceived vibration location moves continuously along the finger.
- **Synchronous vibration:** As a baseline condition, vibration is presented to the finger with constant intensity. Unlike the position-dependent mode, the vibration pattern remains spatially static regardless of the hand’s depth within the object, providing only contact timing cues.

3 USER STUDY

This section describes a user study conducted to evaluate the effects of the proposed visual and haptic feedback on the perception of 3D objects. This experiment was approved by the Ethics Committee of The University of Osaka (No. R7-23), and all participants provided written informed consent prior to their participation.

Table 1: Summary of questionnaire items for subjective assessment.

Category	Item
Body Ownership	1. I felt as if I was looking at my own hand. 2. I felt as if the projected hand was part of my body. 3. I felt as if the projected hand was my hand.
Sense of Agency	4. The projected hand moved according to my intentions, as if it were obeying my will. 5. I felt as if I was controlling the movements of the projected hand. 6. I felt as if I was the one causing the movements I was seeing.
Visual Realism	7. I felt that the visual interaction between the object and the finger was realistic.
Penetration Sensation	8. I felt as if my finger was actually penetrating the 3D object.
Tactile Sensation	9. I felt a sensation of touching the 3D object.
Tactile Realism	10. I felt that the tactile sensation was close to that of a real touch.

3.1 Experimental Design and Participants

Fourteen participants took part in the study (13 males, 1 female; 13 right-handed, 1 left-handed; mean age 22.8 ± 0.7 years). The experiment employed a **3 (Visual Effect) $\times$ 3 (Vibration Effect) within-subjects factorial design**. The independent variables were the Visual Effect (Penetration, Non-penetration, and Warp) and the Vibration Effect (Position-dependent, Synchronous, and None), yielding a total of nine experimental conditions.

3.2 Procedure

In the experimental procedure, participants first experienced all visual effects and haptic stimuli across the nine conditions as a preliminary orientation. Subsequently, they performed a 60-second task followed by a subjective evaluation for each condition. The order of the conditions was randomized to minimize order effects.

During the experiment, participants wore noise-canceling headphones. This served a dual purpose: **masking the mechanical noise of the vibrators** to prevent auditory cues from influencing tactile perception, and **presenting a metronome-like sound** to guide the timing of the point-following task, ensuring consistent movement speed across participants.

The task consisted of the following three 20-second phases designed to evaluate spatial perception and interaction quality:

- 1. Depth Perception:** Following a moving point displayed in front of and behind the 3D object using the projected extended hand to perceive depth relationships.
- 2. Visual Effect Observation:** Moving the physical hand vertically according to instructions while following the point to observe dynamic changes in the visual effects (e.g., masking behavior).
- 3. Interaction:** Continuing the vertical movement of the physical hand while the point was fixed at a position where the 3D object could be touched, confirming the spatial relationship and contact sensation with the object.

After each condition, participants responded to ten items across six categories, as shown in Tab. 1, using a 7-point Likert scale (1: strongly disagree to 7: strongly agree).

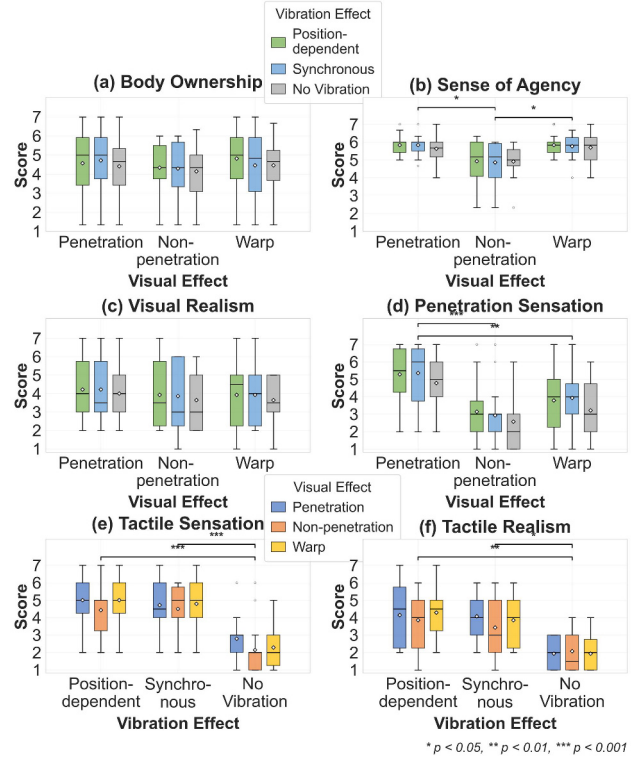


Figure 3: Experimental Results

3.3 Results

The boxplots of the subjective evaluation results are shown in Fig. 3. To clearly illustrate the primary factors influencing each metric, the plots are grouped by their X-axis: (a)–(d) are plotted against the **Visual Effect**, while (e) and (f) are plotted against the **Vibration Effect**. We conducted a two-way repeated-measures analysis of variance (ANOVA) with Visual Effect (3 levels) and Vibration Effect (3 levels) as within-subjects factors. Where significant main effects were found, post-hoc pairwise comparisons were performed using paired t-tests with Bonferroni correction. Effect sizes are reported as partial eta squared (η_p^2).

For the sense of body ownership (Fig. 3(a)), although a main effect of the Visual Effect was observed ($p < .05$), post-hoc tests with Bonferroni correction did not reveal statistically significant differences between specific pairs of conditions. The mean values for each condition ranged from 4.33 to 4.57, remaining within a relatively high and uniform range.

A significant main effect of the Visual Effect was observed for the sense of agency (Fig. 3(b)) ($F(2,26) = 9.20, p < .001, \eta_p^2 = 0.195$). Post-hoc comparisons revealed that the **Penetration** and **Warp** conditions yielded significantly higher scores than the Non-penetration condition ($p < .05$). No significant difference was found between the Penetration and Warp conditions. Subjective reports for the Non-penetration condition included comments that it was “the most natural and realistic in appearance,” yet also highlighted inconveniences such as “losing track of hand position” and being “difficult to operate.” Conversely, while the Penetration condition was described as “physically impossible,” it was positively evaluated for its ease of use and operational continuity.

Regarding visual realism (Fig. 3(c)), no significant main effects or interactions were found for any factor ($p > .05$).

Regarding the penetration sensation (Fig. 3(d)), significant

main effects were found for both Visual ($F(2,26) = 20.00, p < .001, \eta_p^2 = 0.249$) and Vibration ($p < .05$) effects. Post-hoc tests indicated that the **Penetration** visual condition significantly outperformed both the Warp and Non-penetration conditions ($p < .05$), recording the highest mean value of 5.14.

Regarding tactile sensation (Fig. 3(e)) and tactile realism (Fig. 3(f)), the main effect of the Vibration Effect was prominent ($p < .001$; Sensation: $F(2,26) = 25.55, \eta_p^2 = 0.384$; Realism: $F(2,26) = 13.79, \eta_p^2 = 0.319$). The conditions involving vibration (**Position-dependent** and **Synchronous**) received significantly higher evaluations compared to the No Vibration condition ($p < .05$). However, no statistically significant difference was found between Position-dependent and Synchronous vibration modes; subjective reports indicated that many participants found the two types of vibration “almost indistinguishable.”

No significant interaction effects between Visual and Vibration factors were observed for any of the evaluation items ($p > .05$). This suggests that the visual and haptic modalities contributed to the user’s perception without statistically detectable modulation of one by the other in this experimental setup.

3.4 Discussion

The experimental results provided partial support for our hypotheses. Specifically, visual effects significantly influenced the sense of agency (supporting **H1-Agency**) and penetration sensation, while vibration effects significantly enhanced tactile realism (supporting **H2**). However, **H1** was not supported regarding body ownership. The absence of significant interaction effects suggests that visual and haptic modalities contributed additively to the user’s perception in this setup.

3.4.1 Sense of Agency and Visual Causality

The visual effects had a prominent impact on the sense of agency (Fig. 3(b)). The Non-penetration condition, which maintained physical correctness by keeping the hand on the surface, resulted in significantly lower agency scores compared to the Penetration and Warp conditions. Kawabe et al. [3] propose that the sense of agency relies on “action-effect causality”—the perceived link between voluntary movement and visual consequences. In the Non-penetration condition, the projected hand stops at the object’s surface even as the user continues to push forward. This discrepancy disrupts the causal link between motor intention and visual feedback. In contrast, the Penetration and Warp effects, although physically unnatural, maintained visual motion in response to the user’s input, thereby preserving the perceptual causality and sustaining the sense of agency. This finding aligns with interaction research in VR suggesting that functional responsiveness often outweighs physical realism for operational performance [5].

3.4.2 Inhibition of Body Ownership

Contrary to H1, no significant difference was found in body ownership across visual conditions (Fig. 3(a)). We attribute this to the high sensitivity of body ownership to temporal and semantic inconsistencies. First, regarding temporal consistency, Shimada et al. [9] demonstrated that a visuotactile temporal discrepancy exceeding 200–300 ms significantly inhibits the rubber hand illusion (body ownership). While the latency of our system was sufficient to maintain agency (which is generally more robust to delay), it may have exceeded the strict threshold required for ownership, acting as a dominant inhibitory factor across all conditions. Second, regarding semantic consistency, the “penetration” phenomenon creates a cognitive conflict; seeing one’s hand pass through a solid object contradicts real-world physics. This top-down cognitive dissonance may have prevented the brain from fully accepting the projected hand as “one’s own body part,” counteracting the benefits of spatial synchronization.

3.4.3 Role of Haptic Feedback

Regarding tactile presentation, conditions with vibration (Position-dependent and Synchronous) yielded significantly higher tactile realism (Fig. 3(f)) than the No Vibration condition, supporting H2. Although statistical analysis did not reveal a significant difference between Position-dependent (PhS) and Synchronous modes in terms of general realism, the PhS mode was essential for maximizing the “Penetration Sensation” (Fig. 3(d)). Participants reported that while the Synchronous mode felt like a simple collision, the Position-dependent vibration provided a spatial “moving” sensation that matched the visual depth of the hand entering the object. The lack of interaction effects implies that users processed the “timing of contact” (haptics) and the “depth of penetration” (vision) as separate but complementary cues. Therefore, to enhance the bodily sensation in projected hand systems, it is effective to integrate visual effects that prioritize causal motion (Penetration/Warp) with haptic feedback that ensures temporal synchronization, rather than strictly adhering to physical consistency.

4 CONCLUSION

We developed a projection-based hand interaction system for 3D objects that integrates visual masking effects with position-dependent vibration using PhS. Our user study revealed a functional independence between sensory modalities: visual penetration effects significantly enhanced the sense of agency by resolving depth discrepancies, while haptic feedback independently contributed to tactile realism. We conclude that prioritizing perceptual causality over physical consistency in visual feedback, combined with synchronized haptic cues, is an effective strategy for maintaining the user’s bodily sensation during 3D interaction.

In future work, we will address the system latency to potentially enhance body ownership, which was not observed in this study. We also plan to explore novel visual rendering techniques to improve visual realism and extend the system to convey internal object states and subtle differences in micro-textures based on high-fidelity vibration information.

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